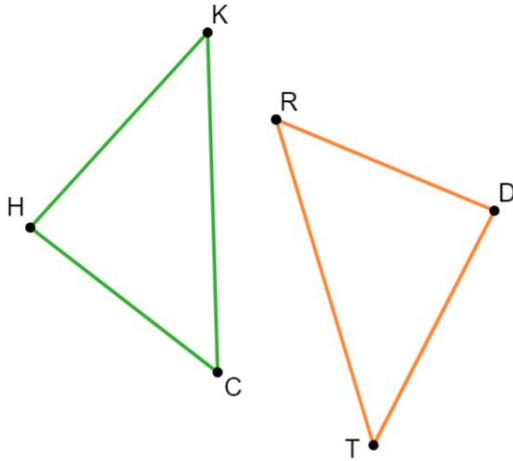




5.4.5 Wrap-Up: Cool Down

The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$.



- Complete each congruence statement.

$$\overline{KC} \cong \underline{\hspace{2cm}}$$

$$\overline{CH} \cong \underline{\hspace{2cm}}$$

$$\angle RTD \cong \underline{\hspace{2cm}}$$

$$\angle RDT \cong \underline{\hspace{2cm}}$$

- Becky told her teacher that there was a typo in the first sentence. She circled the typo.

The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$.

Write a note to Becky explaining why this is NOT a typo.

Unit 5, Lesson 5: Drawing Conclusions from Given Information



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can draw and mark figures using given information.
- ☐ I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from given information.



5.5.1 Warm-Up: Working Through Challenges

Teri takes dance in school, but learning how to tap dance was a new and overwhelming experience. She said that after her teacher introduced tap, “It was really hard for me to learn ... It just looked like a lot of work and looked like it took a lot of time to learn, and it was really high tempo.”

1. What is something that felt overwhelming the first time you were learning it?

Teri’s teacher slowed down the moves to show her the structure and recommended that she practice each step before trying to put it all together. She connected this experience to learning other things that are difficult, saying, “The more you practice a certain skill that you need, the better you become at it, and the easier it becomes.”

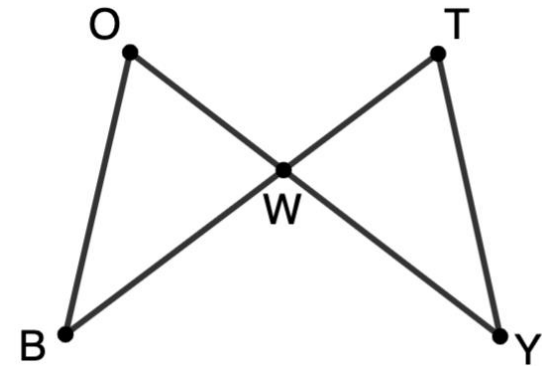
2. Describe a time when practicing something that was challenging for you helped you become better at it.



5.5.2 Collaboration: Using Given Information to Mark Figures

When given information about geometric figures, it is helpful to mark the figures.

1. Triangles OBW and WTY are shown.



Josue and Abby marked the figure in different ways given $\overline{OW} \cong \overline{TW}$, $\overline{OB} \cong \overline{TY}$, and $\angle OBW \cong \angle TYW$ as shown.

Josue's Markings	Abby's Markings

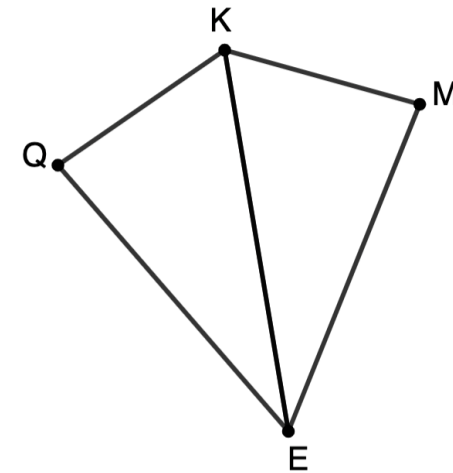
- a. Discuss with your partner what you notice about how Josue and Abby each marked the figure.

- b. Complete the table to describe similarities and differences in the way each student marked the figure.

How were their markings similar?	How were their markings different?

- c. Josue wondered if there was enough information given to conclude $\triangle OWB \cong \triangle TWY$. Work with your partner to explain whether there is enough information to conclude $\triangle OWB \cong \triangle TWY$.
- d. Abby said she does not think there is enough given information to conclude $\triangle OWB \cong \triangle TWY$, but she said there is enough information to conclude $\angle OWB \cong \angle TWY$. What definition is Abby using to draw this conclusion?
- e. Add markings to Abby's figure to represent her conclusion that $\angle OWB \cong \angle TWY$.
- f. Work with your partner to explain whether there is now enough information to conclude $\triangle OWB \cong \triangle TWY$.

2. Mark polygon $KQEM$ using the given information.
 $\overline{EQ} \cong \overline{EM}$ and $\angle KEQ \cong \angle KEM$



3. Compare your markings with your partner. Make any corrections if necessary.



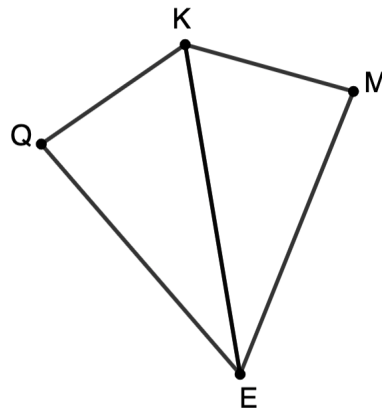
5.5.3 Guided Instruction: Drawing Conclusions From Given Information

Given information, definitions, properties, postulates, and theorems can be used to draw conclusions about figures.

- What are some definitions, properties, postulates, and theorems related to triangles and angles learned so far this year?

Definitions	
Properties	
Theorems and Postulates	

Polygon $KQEM$, where $\overline{QK} \cong \overline{KM}$ and $\overline{EQ} \cong \overline{EM}$, is shown. $\overline{KE}$ is drawn inside polygon $KQEM$.



- Mark the figure using the given information.
- Discuss any conclusions that can be drawn about the figure with the information given.

Using the given information, Hilary concludes $\triangle QEK \cong \triangle MEK$.

- Complete the statement to explain how Hilary is able to draw this conclusion.

Hilary noticed

☐ 1
☐ 2
☐ 3

 corresponding

☐ side(s)
☐ angle(s)

 of

$\triangle QEK$ and $\triangle MEK$ are congruent based on the given information.

Both triangles also share a common

☐ side,
☐ angle,

which is congruent to itself because of the

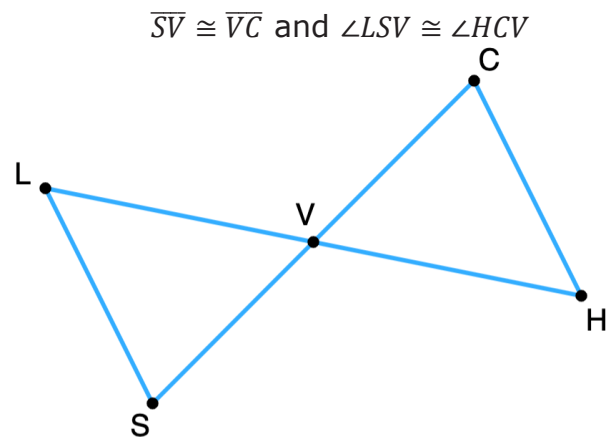
☐ reflexive
☐ transitive

 property.

She can conclude $\triangle QEK \cong \triangle MEK$ because of

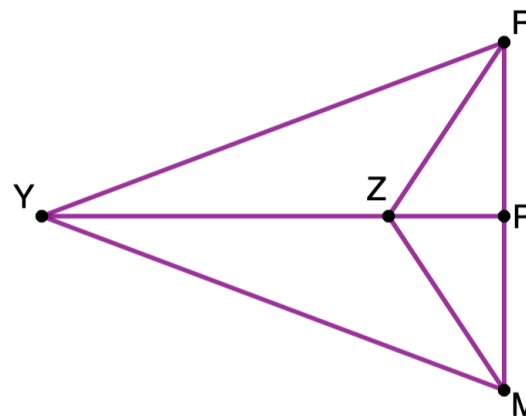
☐ SSS.
☐ SAS.
☐ ASA.
☐ AAS.
☐ HL.

5. Triangle LVS and triangle CVH are shown.



- a. What conclusions can be drawn from the given information?
- b. What congruence postulate or theorem is the reason these conclusions can be made?
- c. Julia says she can conclude $\overline{CH} \cong \overline{SL}$ and $\angle SLV \cong \angle CHV$. Explain why Julia is able to draw these conclusions.

6. Triangle RMY is shown. $\angle YMZ \cong \angle YRZ$ and $\angle RZY \cong \angle MZY$.



Work with your partner to complete the statements based on the given information.

- a. I can conclude ...
- b. I know this because ...

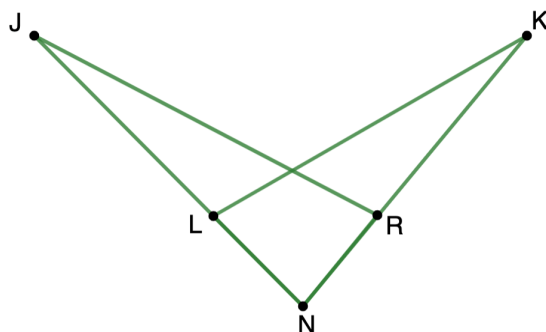


5.5.4 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

Dalton saw the following question with the polygon shown in an old geometry textbook.

"If $\angle NKL \cong \angle NJR$ and $\overline{KL} \cong \overline{JR}$, what else do you know so that you can show that $\triangle RJN \cong \triangle LKN$ using ASA?"



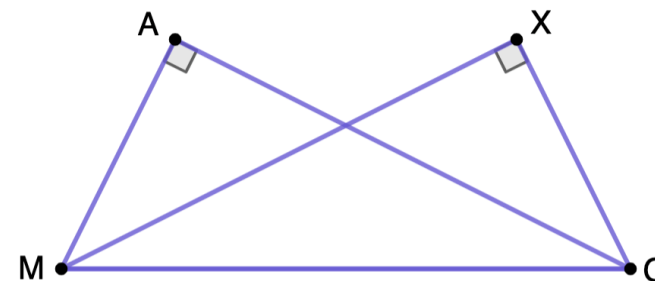
He knew that both triangles share $\angle N$. Therefore, by knowing that $\angle N \cong \angle N$ by the reflexive property of congruence, he concluded that he could use ASA to show that $\triangle RJN \cong \triangle LKN$.

1. Draw the triangles so that they do not overlap. Then, mark the triangles with the information given from the textbook.
2. Explain why Dalton can or cannot use ASA to show that $\triangle RJN \cong \triangle LKN$.



5.5.5 Try It: Drawing Conclusions From Given Information

1. Triangle MAO and triangle XOM are shown, where $\angle MAO$ and $\angle OXM$ are right angles and $\overline{MA} \cong \overline{XO}$.



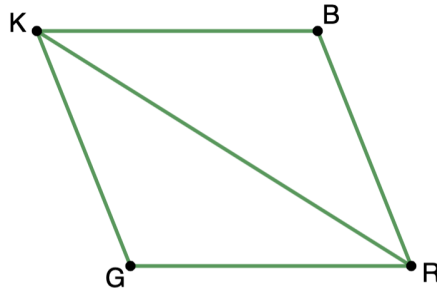
Complete each statement based on the given information.

- a. I can conclude that ...
- b. I know this because ...



5.5.6 Wrap-Up: Ticket Out the Door

Quadrilateral $BRGK$ is shown.



Complete each statement based on the given information.

1. Given that $\overline{KB} \cong \overline{GR}$ and $\overline{KG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
2. Given that $\overline{KB} \cong \overline{KG}$ and $\overline{RG} \cong \overline{BR}$,
 - a. I can conclude that ...
 - b. I know this because ...
3. Given that $\overline{KG} \parallel \overline{BR}$ and $\overline{KB} \parallel \overline{GR}$,
 - a. I can conclude that ...
 - b. I know this because ...

Unit 5, Lesson 6: Proving Triangles Congruent



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

- ☐ I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.